The Physical Tether Theorem: A Rigorous Formal Synthesis of Representational Measurement Theory, Operationalism, and Classical Thermodynamics

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April 5, 2026

Abstract

The necessary and sufficient conditions for an abstract construct to serve as a useful probe of physical reality follow directly from the first principles of classical science. The **Physical Tether Theorem** (Theorem 1) states them: an abstract construct A is a scientifically valid probe of physical reality PR if and only if A is tethered to PR by a unique induced quotient-valued operational mapping (equivalently, unique up to empirical-content symmetries). This theorem is a direct mathematical consequence of two long-established foundations of classical science, the representational measurement theory of Krantz, Luce, Suppes & Tversky (1971–1990) [10, 15, 16] and the operationalist criterion of Bridgman (1927) [2], both resting on the classical-realist principle that every physical quantity has a single objective value at every physical state. The **Non-Aggregation Theorem for Intensive Variables** (Theorem 2), proved as a direct structural consequence of the zeroth law and the Clausius form of the second law of thermodynamics, states that on any macroscopic non-equilibrium system with non-constant intensive field, no scalar can serve as a value of that intensive variable for the system in the operational sense of classical thermodynamics. The **Thermodynamic Application Theorem** (Theorem 3) follows: if a proposed global construct on a non-equilibrium system requires choosing an aggregation operator for local intensive quantities, the construct is untethered and therefore not a scientifically valid probe of physical reality. Applied to the GMST construct (as used by the IPCC, NASA, and others), the result is that GMST is untethered and physically meaningless. No new axiom is proposed. The Physical Tether Theorem (Theorem 1), the Non-Aggregation Theorem (Theorem 2), the Thermodynamic Application Theorem (Theorem 3), and the physical meaninglessness of GMST are direct consequences of classical science itself.

Key Result: The GMST construct is physically meaningless.

Keywords: Physical Tether Theorem, Non-Aggregation Theorem, representational measurement theory, operationalism, classical thermodynamics, GMST, intensive quantities, non-equilibrium systems, admissible operational mapping, aggregation operator, scientific validity

1 Motivation and Scope

Representational measurement theory distinguishes empirical relational structure from numerical representation and proves uniqueness theorems only up to admissible transformations [10, 15, 16]. Bridgman's operationalism requires that physical meaning be tied to physically specified operations [2]. The Physical Tether Theorem (Theorem 1) synthesizes these two traditions into a single formal equivalence between scientific validity and tetheredness. Classical thermodynamics, as axiomatized by Callen (1985) [3], distinguishes local intensive quantities from additive extensive quantities and sharply limits what may be combined by law-governed operations; the Non-Aggregation Theorem (Theorem 2) and the Thermodynamic Application Theorem (Theorem 3) derived from it rest only on the zeroth law and the Clausius form of the second law.

The Physical Tether Theorem (Theorem 1) is *meta-level*: it does not by itself establish that any particular construct is untethered. It proves that an abstract construct functions as a scientifically valid probe of physical reality if and only if the operational route from physical states to the reported construct is uniquely fixed up to empirically irrelevant symmetry. A separate physical argument is then required in any concrete application. For intensive thermodynamic variables on non-equilibrium systems, that separate argument is supplied by the Non-Aggregation Theorem (Theorem 2, Section 4), which rests on the zeroth law and the Clausius form of the second law alone.

2 Formal Framework

Foundational Principle (Classical Realism). All definitions in this framework operate within classical science and inherit its foundational commitment to classical realism: every objective physical quantity is single-valued at every physical state, independent of human description or choice. This principle is Bridgman's (1927) operationalist demand transposed into the language of representational measurement theory (Krantz, Luce, Suppes, and Tversky, 1971–1990): every physical concept must be tied to a unique set of operations fixing one objective value per state, up to empirical-content equivalence. The formal statement of this principle is given in Remark 4 below. It is invoked by name throughout the definitions that follow.

Definition 1 (Physical state space)**.** Let PR denote physical reality and let $\mathcal{S} \subseteq \mathsf{PR}$ be a specified class of objective physical states. The elements of $\mathcal{S}$ are the states over which the proposed construct is intended to probe empirical reality.

Definition 2 (Representation space)**.** For a given abstract construct A, let R_A be the representation space in which reports of A take values. Depending on context, R_A may be a numerical space, a function space, an ordered set, or some other mathematically specified target.

Definition 3 (Admissible operational mapping)**.** An *admissible operational mapping* for construct A is a map

$$\phi : \mathcal{S} \to R_A$$

satisfying both:

(i) *[Bridgman operationalism.]* ϕ is fully specified by concrete physical operations permitted by the physical laws governing $\mathcal{S}$; and

(ii) *[Classical realism / single-valuedness, Remark 4.]* every aggregation operator (if any) required by the construction of A is thereby uniquely determined by those physical laws, up to empirical-content equivalence in the sense of representational measurement theory (Krantz, Luce, Suppes, and Tversky, 1971–1990).

Human conventions or arbitrary choices of aggregator that are not dictated by the governing physical axioms do not satisfy the single-valuedness required by classical realism and therefore cannot yield a real physical quantity. Whether the admissibility criterion is met for any particular construct A on any particular state class $\mathcal{S}$ is a question answered by the physical theory governing $\mathcal{S}$; it is not settled by definition alone.

Remark 1 (Grounding of Definition 3 in classical realism)**.** Clause (ii) of Definition 3 is not an additional postulate. It is the direct consequence of the classical realism requirement (Remark 4), that every physical property have a single value per physical state, applied to the operational mapping specified in (i). Any construct whose reported value depends on the aggregator chosen by the person computing it violates single-valuedness by construction and by that very fact fails to correspond to any real physical quantity. The uniqueness demand in (ii) is therefore inherited from the shared foundational axioms of all classical science (realism, operationalism, representational uniqueness). It is not imposed by this definition and cannot be circumvented by rejecting this definition. An objection to (ii) is an objection to classical realism itself.

Remark 2. The term "admissible" is intentionally stronger than "mathematically definable." A map is admissible only when each physically meaningful step in the construction is permitted by the underlying physical theory and the operational protocol, not merely by algebraic convenience.

Definition 4 (Empirical-content symmetry and quotient space)**.** Let $\sim_A$ be an equivalence relation on R_A representing the symmetries that preserve empirical content for the construct A; for example, admissible unit changes or other representation changes that leave all physical content unchanged. Define the quotient space

$$Q_A := R_A/\!\sim_A,$$

and let

$$q_A : R_A \to Q_A$$

be the quotient map.

Definition 5 (Equivalence of admissible mappings)**.** For $\phi, \psi \in \mathcal{M}_A$, define

$$\phi \approx_A \psi \quad \Longleftrightarrow \quad q_A \circ \phi = q_A \circ \psi.$$

Thus two admissible mappings are equivalent exactly when they induce the same empirical-content map from $\mathcal{S}$ to Q_A.

Definition 6 (Induced quotient map)**.** Each admissible mapping $\phi \in \mathcal{M}_A$ (a map $\mathcal{S} \to R_A$) induces the quotient-valued map

$$\bar{\phi} := q_A \circ \phi : \mathcal{S} \to Q_A.$$

The family of all such induced quotient maps (maps $\mathcal{S} \to Q_A$) is denoted

$$\overline{\mathcal{M}_A} := \{\bar{\phi} \mid \phi \in \mathcal{M}_A\}.$$

(Note that $\mathcal{M}_A$ and $\overline{\mathcal{M}_A}$ are formally distinct families because their codomains differ.)

Definition 7 (Admissible predicate)**.** An *admissible predicate* (representing an empirical claim) for A is a map

$$P : Q_A \to \{0, 1\}.$$

Its truth value at physical state $s \in \mathcal{S}$ under operational mapping $\phi \in \mathcal{M}_A$ is

$$P\Big(\bar{\phi}(s)\Big) \in \{0, 1\}.$$

Remark 3. Every Boolean function on Q_A qualifies as an admissible predicate. This is not a liberal assumption but a necessary one: since Q_A is the quotient of a representation space by empirical-content symmetries, distinct elements of Q_A are by construction empirically distinguishable, and the indicator function that separates any two distinct elements is therefore a legitimate empirical claim. The broad class ensures the Separation Lemma (Lemma 1) holds for every conceivable empirical distinction without exception.

Definition 8 (Useful probe)**.** The abstract construct A is a *useful probe of physical reality on* $\mathcal{S}$, written $U(A, \mathsf{PR}; \mathcal{S})$, if and only if:

(i) $\mathcal{M}_A \neq \varnothing$, and

(ii) for every $s \in \mathcal{S}$, for every admissible predicate $P : Q_A \to \{0, 1\}$, and for every $\phi, \psi \in \mathcal{M}_A$,

$$P\Big(\bar{\phi}(s)\Big) = P\Big(\bar{\psi}(s)\Big).$$

Equivalently: every admissible empirical truth value attached to A is determined solely by the physical state and not by admissible mapping choice. This strong condition ensures that the reported value of A is independent of arbitrary human conventions in the operational procedure; if it were not, the construct would reflect the choice of mapping rather than physical reality itself.

Definition 9 (Scientifically valid probe)**.** The abstract construct A is a *scientifically valid probe of physical reality on* $\mathcal{S}$, written $V(A, \mathsf{PR}; \mathcal{S})$, if and only if it is a useful probe, i.e.,

$$V(A, \mathsf{PR}; \mathcal{S}) \iff U(A, \mathsf{PR}; \mathcal{S}).$$

Definition 10 (Tethered)**.** The abstract construct A is *tethered to physical reality on* $\mathcal{S}$, written $T(A, \mathsf{PR}; \mathcal{S})$, if and only if

$$\left|\overline{\mathcal{M}_A}\right| = 1.$$

Equivalently: there exists exactly one quotient-valued admissible operational mapping from $\mathcal{S}$ to Q_A.

Each definition above is the exact mathematical embodiment of a first principle of classical science.

Remark 4 (Classical Realism Requirement)**.** In classical physics, classical realism (and classical determinism) requires that every objective physical quantity be single-valued: for any given physical state $s \in \mathcal{S}$, the quantity takes exactly one value in the real world, independent of human description or choice.

This single-valuedness is the direct mathematical transcription of Bridgman's (1927) demand that every physical concept be synonymous with a unique set of operations fixing one objective value.

Any construct A that admits infinitely many non-equivalent admissible representations (i.e., $|\overline{\mathcal{M}_A}| > 1$) or none at all ($|\overline{\mathcal{M}_A}| = 0$) thereby violates single-valuedness and cannot correspond to any real physical quantity.

The GMST construct is formed by an arbitrary human choice of aggregator among infinitely many non-equivalent possibilities (arithmetic mean, root-mean-square, geometric mean, harmonic mean, etc.), with no selection rule supplied by the axioms of classical thermodynamics. It therefore fails single-valuedness and is not a real physical quantity.

3 The Physical Tether Theorem

Lemma 1 (Separation by a quotient map)**.** *For any distinct $x, y \in Q_A$, there exists an admissible predicate $P : Q_A \to \{0, 1\}$ such that $P(x) \neq P(y)$.*

Proof. Define $P(z) = 1$ if $z = x$, and $P(z) = 0$ otherwise. This is a Boolean map on Q_A and hence admissible. Then $P(x) = 1 \neq 0 = P(y)$. □

Theorem 1 (Physical Tether Theorem)**.** *An abstract construct A is a useful probe of physical reality on $\mathcal{S}$ if and only if it is tethered to* PR *on $\mathcal{S}$. Equivalently, $V(A, \mathsf{PR}; \mathcal{S}) \iff T(A, \mathsf{PR}; \mathcal{S})$.*

Proof. ($\Rightarrow$) Suppose $U(A, \mathsf{PR}; \mathcal{S})$. Then $\mathcal{M}_A \neq \varnothing$ and for all $s \in \mathcal{S}$, all admissible P, all $\phi, \psi \in \mathcal{M}_A$, $P(\bar{\phi}(s)) = P(\bar{\psi}(s))$. Suppose for contradiction that $|\overline{\mathcal{M}_A}| \neq 1$. If $|\overline{\mathcal{M}_A}| = 0$, then $\mathcal{M}_A = \varnothing$, contradicting U. If $|\overline{\mathcal{M}_A}| \geq 2$, then there exist $\phi, \psi \in \mathcal{M}_A$ and $s \in \mathcal{S}$ with $\bar{\phi}(s) \neq \bar{\psi}(s)$. By the Separation Lemma (Lemma 1), there exists admissible P with $P(\bar{\phi}(s)) \neq P(\bar{\psi}(s))$, contradicting U. Thus $|\overline{\mathcal{M}_A}| = 1$, so $T(A, \mathsf{PR}; \mathcal{S})$.

($\Leftarrow$) Suppose $T(A, \mathsf{PR}; \mathcal{S})$. Then $|\overline{\mathcal{M}_A}| = 1$, so $\mathcal{M}_A \neq \varnothing$ and all $\bar{\phi}$ are identical. For any admissible P and $s \in \mathcal{S}$, $P(\bar{\phi}(s))$ is the same for all $\phi \in \mathcal{M}_A$. Thus $U(A, \mathsf{PR}; \mathcal{S})$, so $V(A, \mathsf{PR}; \mathcal{S})$. □

Remark 5 (Universality Across Formalizations of Classical Thermodynamics)**.** The Physical Tether Theorem (Theorem 1) expresses a core requirement of classical science itself, not a feature peculiar to any single formal language. Classical realism (unique objective value per physical state) together with Bridgman's operationalism underpins *every* formalization of classical science. The Non-Aggregation Theorem (Theorem 2, Section 4) proves that no scalar can serve as a value of a non-constant intensive field on a non-equilibrium system in the operational sense of classical thermodynamics, using only the zeroth law and the Clausius form of the second law. The same conclusion follows, *mutatis mutandis*, for every other intensive conjugate variable paired with its own equilibrium condition (pressure under mechanical equilibrium, chemical potential under chemical equilibrium, electric potential under electrostatic equilibrium, and so on). The zeroth law and the Clausius form of the second law are definitional of classical thermodynamics itself; any formalization lacking either is not a formalization of classical thermodynamics. The Non-Aggregation Theorem, the Thermodynamic Application Theorem (Theorem 3), and the untetheredness of GMST therefore hold in *every* formalization of classical thermodynamics without exception, including Callen [3], Carathéodory (1909) [5], Landau & Lifshitz (1980) [11], Lieb & Yngvason (1999) [14], Giles (1964) [7], Truesdell (1984) [19], Lavis & Frigg (2025) [12], and every variant of rational thermodynamics.

Consequently, any abstract construct A that is not tethered has zero physical meaning in classical physics.

4 The Non-Aggregation Theorem and Its Application

Assumption 1 (Local intensive field on non-equilibrium system)**.** A is a proposed global construct on a non-equilibrium system $\mathcal{S}$ formed by combining local intensive values.

Assumption 2 (Aggregator-dependent global construct)**.** The construction of A requires choosing an aggregation operator $F \in \mathcal{F}$ from a family $\mathcal{F}$ of possible aggregators.

Theorem 2 (Non-Aggregation Theorem for Intensive Variables)**.** *Let $\mathcal{S}$ be a macroscopic non-equilibrium system with non-constant local temperature field $T(\cdot)$.*

(a) ***Structural impossibility.*** *No scalar T^* can serve as a temperature of $\mathcal{S}$ in the operational sense of classical thermodynamics. Equivalently, the second law supplies no aggregation operator that maps the local temperature field to a temperature of $\mathcal{S}$. The same holds,* mutatis mutandis*, for every intensive conjugate variable (pressure, chemical potential, . . .) paired with its own equilibrium condition and its own second-law statement.*

(b) ***Consequence.*** *For any proposed global intensive construct A on $\mathcal{S}$, no map $\phi : \mathcal{S} \to R_A$ satisfies the admissibility criterion of Definition 3. Consequently $\mathcal{M}_A = \varnothing$.*

Proof. Part (a). Suppose, for contradiction, that a scalar T^* is a temperature of $\mathcal{S}$ in the operational sense of classical thermodynamics. Then $\mathcal{S}$ must be in thermal equilibrium at T^*, so that thermal contact with a reference body R held uniformly at T^* produces no spontaneous transfer of energy as heat at any point of the interface. By the Clausius form

of the second law, the absence of spontaneous transfer of energy as heat across a point of thermal contact requires equal temperatures on the two sides. Hence $T(x) = T^*$ at every interface point x.

Because $T(\cdot)$ is non-constant, there exists a point $y \in \mathcal{S}$ with $\nabla T(y) \neq 0$. Consider a thermal-contact interface that intersects a neighborhood of y. Any such neighborhood has non-constant temperature, so no single value T^* can equal $T(x)$ throughout the neighborhood. Contradiction.

Therefore no such T^* exists, and no aggregation operator can produce one. The identical reasoning, with the appropriate equilibrium condition and second-law statement, applies to every other intensive thermodynamic variable (pressure under mechanical equilibrium, chemical potential under chemical equilibrium, and so on).

Part (b). By Definition 3, a mapping $\phi : \mathcal{S} \to R_A$ is admissible only if every aggregation operator it employs is uniquely selected by the governing physical laws up to empirical-content equivalence. Part (a) proves that no such selection exists for any intensive aggregation over $\mathcal{S}$. Therefore no ϕ satisfies the admissibility criterion, so $\mathcal{M}_A = \varnothing$. □

Theorem 3 (Thermodynamic Application Theorem)**.** *Under Assumptions 1 and 2, and the Non-Aggregation Theorem (Theorem 2), if a proposed global construct A requires choosing an aggregation operator for local intensive quantities, then A is not tethered to physical reality on $\mathcal{S}$. Consequently,*

$$\neg T(A, \mathsf{PR}; \mathcal{S}) \quad \textit{and hence} \quad \neg V(A, \mathsf{PR}; \mathcal{S}).$$

Proof. By the Non-Aggregation Theorem (Theorem 2), part (b), $\mathcal{M}_A = \varnothing$. By the Physical Tether Theorem (Theorem 1), $\neg T(A, \mathsf{PR}; \mathcal{S})$, hence $\neg V(A, \mathsf{PR}; \mathcal{S})$. □

Corollary 1 (Conditional intensive-field corollary)**.** *Under the Non-Aggregation Theorem (Theorem 2), let A be a proposed global construct formed by combining local intensive values across a non-equilibrium system. Then A is untethered and therefore not a scientifically valid probe of physical reality.*

Proof. Immediate from the preceding theorem. □

Remark 6 (Concrete illustration)**.** Consider a one-dimensional rod of length $L = 1$ with steady-state heat conduction (non-equilibrium) and linear temperature field

$$T(x) = 300 + 100x \quad \text{(K)}, \quad x \in [0, 1].$$

Possible global-temperature constructs require an aggregator F:

- Arithmetic mean: $\frac{1}{L}\int_0^L T(x)\,dx = 350$
- Root-mean-square: $\sqrt{\frac{1}{L}\int_0^L T(x)^2\,dx} \approx 351.19$

- Geometric mean: $\exp\left(\frac{1}{L}\int_0^L \ln T(x)\,dx\right) \approx 348.8$

By the Non-Aggregation Theorem (Theorem 2), no scalar can serve as a temperature of the rod. The three numbers above are each a legitimate function of $T(x)$; none is a temperature of the rod. The field $T(x)$ itself is the complete thermodynamic description. The unit tag "K" on the local field is physical; the same tag carried through an aggregator yields a dimensionally consistent number without a physical referent.

4.1 Applications to Global Climate Constructs

Remark 7 (GMST as calculated by the IPCC, NASA, and others)**.** Global Mean Surface Temperature (GMST) is calculated by the IPCC, NASA, and other groups by aggregating a local intensive temperature field over a non-equilibrium system (the surface air and ocean water of the Earth) using an area-weighted arithmetic mean, as detailed by IPCC (2018) [9] and Lenssen et al. (2019) [13]. By the Non-Aggregation Theorem (Theorem 2) and Definition 3, no admissible operational mapping exists for this construct. Therefore $\mathcal{M}_A = \varnothing$, so $\neg T(\text{GMST}, \mathsf{PR}; \mathcal{S})$ and hence $\neg V(\text{GMST}, \mathsf{PR}; \mathcal{S})$. By the Classical Realism Requirement (Remark 4), the GMST construct is not a real physical quantity.

It follows that the GMST construct is untethered and therefore physically meaningless in the domain of classical physics. Equivalently, GMST is not a scientifically valid probe of physical reality. This conclusion follows directly from the zeroth law and the Clausius form of the second law, with no additional postulates.

The fact that any particular group (IPCC, NASA, etc.) has codified one specific aggregator does not render that choice admissible; admissibility requires unique selection by the physical laws themselves (see Definition 3).

Remark 8 (Standard defenses rebutted)**.** The Non-Aggregation Theorem (Theorem 2) rules out every scalar aggregated from a non-constant intensive field on a non-equilibrium system as a value of that intensive variable for the system. By construction, different aggregators yield different numbers from the same local field, so the scalar is not an extensive quantity either. It is therefore not energy, not internal energy, not a proxy for internal energy, not an indicator of internal energy change, not an approximation of any physical quantity. It is a number, not a physical quantity. Classical thermodynamics supplies no admissible operational mapping that attaches it to any physical property of the system.

The effective emitting temperature construct T_{eff} is a total fiction unrelated to any physical quantity of the Earth system. It is computed by inverting the Stefan–Boltzmann blackbody law applied to a globally averaged outgoing longwave flux. The Earth is not a blackbody: it has no uniform surface, no uniform emissivity, no uniform temperature, and no equilibrium. The outgoing longwave flux itself cannot be measured accurately enough even to resolve the fictitious number, let alone to resolve any alleged change in it over time. T_{eff} is not a temperature of the Earth, not a temperature of its surface, not a temperature of its atmosphere, not a temperature of anything. It is a Stefan–Boltzmann inversion of an average, a number without a physical referent.

Remark 9 (Non-permission is a theorem of classical thermodynamics, not a postulate)**.** The non-permission of intensive aggregation on a macroscopic non-equilibrium system is

not an added postulate. It is the content of the Non-Aggregation Theorem (Theorem 2), which is proved from the zeroth law and the Clausius form of the second law alone. Every formalization of classical thermodynamics therefore yields the same non-permission.

5 Extensive versus Intensive Quantities: The Source of Non-Uniqueness

Classical thermodynamics defines two mutually exclusive types of quantity: extensive and intensive. Extensive quantities are additive over subsystems, and conservation principles or state-function structure select a unique combination law, typically integration or summation over the system. By contrast, an intensive quantity is defined locally and possesses no law-selected additive combination rule over heterogeneous non-equilibrium systems; by the Non-Aggregation Theorem (Theorem 2) the impossibility is structural, not accidental.

This structural difference is precisely what the Non-Aggregation Theorem (Theorem 2) rules out at the level of first principles. Every canonical formalization of classical thermodynamics takes the additivity of extensives as definitional, and takes the zeroth law and the Clausius form of the second law as its universal physical laws. All three are non-negotiable foundations of the subject; any formalization that rejects any of them has stopped being classical thermodynamics. The Non-Aggregation Theorem (Theorem 2) uses all three. Extensive quantities therefore have an aggregation rule by definition: addition over disjoint subsystems or integration over continuous fields. Intensive quantities have none, because the zeroth law together with the second law requires that any scalar presented as a value of an intensive variable for a system be the value on which the system equilibrates with a reference body held at that scalar, and no such scalar exists for a system with a non-constant intensive field. Essex et al. (2007) [6] argued that "the burden to demonstrate the utility of a statistic as an index must lay with those who propose it as one." The Non-Aggregation Theorem (Theorem 2) is stronger: no such global intensive scalar exists for a non-equilibrium system. The question is closed.

6 Relation to Foundational Literature

The content of this paper is not a novel synthesis but a precise mathematical formalization of long-established first principles of classical science. The Physical Tether Theorem (Theorem 1) rests on two of them: the representational measurement theory of Krantz, Luce, Suppes & Tversky [10, 15, 16], which fixes numerical representation only up to admissible transformations, and the operationalist criterion of Bridgman (1927) [2], which requires that every concept be defined by concrete operations; both rest on classical realism, which supplies the uniqueness of physical properties (one reality, one value). The Non-Aggregation Theorem (Theorem 2) and the Thermodynamic Application Theorem (Theorem 3) rest on two additional first principles drawn from classical thermodynamics: the zeroth law and the Clausius form of the second law. The definitions and theorems of this paper are the direct mathematical transcription of these centuries-old principles. The alignment with earlier literature is therefore exact:

- **Helmholtz (1887) [8]:** measurement as preservation of empirical structure under

representation;

- **Campbell (1920) [4]:** the distinction between fundamental and derived measurement;
- **Bridgman (1927) [2]:** the demand that physical meaning be tied to specified operations;
- **Krantz, Luce, Suppes, and Tversky (1971–1990) [10, 15, 16]:** representation and uniqueness theorems, including admissible transformation classes;
- **Callen (1985) [3]:** the formal role of state variables, the distinction between intensive and extensive quantities, and the limits imposed by thermodynamic structure.

For recent critical analyses of the scope of representational measurement theory, in particular the central role of uniqueness theorems in constraining admissible numerical assignments and the formal requirements for structure-preserving mappings, see Tal (2021) [18] and Vessonen (2021) [20]. For a philosophical analysis of thermodynamics as a resource-relative theory, consistent with the operationalism of Definition 3, see Robertson & Prunkl (2023) [17].

The equivalence $U(A, \mathsf{PR}; \mathcal{S}) \iff T(A, \mathsf{PR}; \mathcal{S})$ (and hence $V(A, \mathsf{PR}; \mathcal{S}) \iff T(A, \mathsf{PR}; \mathcal{S})$) is a direct consequence of the uniqueness-up-to-admissible-transformations theorems in representational measurement theory together with Bridgman's operationalism. The present contribution sharpens this equivalence to the level of admissible empirical predicates and integrates it explicitly with the thermodynamic Non-Aggregation Theorem (Theorem 2), which is itself a direct consequence of the zeroth and second laws.

7 Discussion

The Physical Tether Theorem (Theorem 1) is exact: an abstract construct is a scientifically valid probe of physical reality if and only if it is tethered. Consequently, any global intensive construct whose construction requires an aggregation operator is scientifically invalid, because the Non-Aggregation Theorem (Theorem 2) rules out every such operator from the zeroth and second laws alone, and the Thermodynamic Application Theorem (Theorem 3) then forces untetheredness.

This precision matters. Informal statements of the idea can overreach by jumping directly from "non-unique construction" to sweeping physical conclusions without making explicit the representation space, the symmetry quotient, the class of admissible mappings, or the predicate-level notion of empirical truth. The present framework closes those gaps. The thermodynamic premise is a theorem of classical thermodynamics proved from two of its own universally accepted statements.

8 Conclusion

The Physical Tether Theorem (Theorem 1) is an absolute equivalence of classical science: an abstract construct is a scientifically valid probe of physical reality if and only if it is tethered by a unique (up to empirical-content symmetries) admissible operational mapping.

The Non-Aggregation Theorem (Theorem 2) is an absolute structural impossibility within classical thermodynamics: on any macroscopic non-equilibrium system with non-constant intensive field, no scalar can serve as a value of that intensive variable for the system. The Thermodynamic Application Theorem (Theorem 3), together with Corollary 1, forces the GMST construct (area-weighted arithmetic mean of local surface temperatures, as used by the IPCC, NASA, and others) out of the category of real physical quantities. It is formed by an arbitrary human choice among infinitely many non-equivalent aggregators, all of which are ruled out at once by the Non-Aggregation Theorem (Theorem 2).

All of these are direct, unconditional results of classical science itself.

Funding

No funds, grants, or other support was received.

Acknowledgements

The author was inspired to pursue this line of inquiry by the foundational work of Christopher Essex, in particular Essex et al. (2007) [6] and Andresen & Essex (2020) [1], as well as through personal meetings, discussions, and viewing several public videos of his talks on the subject. The author used Grok 4.2–4.7 (xAI) and Claude Opus 4.7 (Anthropic) for assistance with formalization, drafting refinement, and proofreading. All substantive claims, interpretations, and final wording were reviewed, edited, and approved by the author.

This article was prepared in the author's personal capacity. The views and conclusions expressed are solely those of the author and do not necessarily reflect the views of any affiliated institution or organization.

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